

Dense Retriever Optimization: Complete Primary Result Fragments

Contrast	Mean	95% CI	Decision
Muon - AdamW	+0.0016	[-0.0108, +0.0139]	inconclusive
NorMuon - AdamW	+0.0021	[-0.0089, +0.0130]	inconclusive
NorMuon - Muon	+0.0005	[-0.0020, +0.0030]	inconclusive

Table 1: Final-stage retrieval over 14 decontaminated BEIR tasks, with simultaneous paired-task intervals for all three optimizer contrasts.

1 Optimizer Effects on Dense Retrieval

The primary estimand averages all four declared rates within optimizer from a common model, dataset, training schedule, and evaluation suite.

Primary comparison. Across the four-rate surface, Muon versus AdamW is inconclusive at +0.0016 [-0.0108, +0.0139] nDCG@10, and NorMuon versus AdamW is inconclusive at +0.0021 [-0.0089, +0.0130]. Validation-selected comparisons are secondary: Muon versus AdamW inconclusive at +0.0032 [-0.0016, +0.0080]; NorMuon versus AdamW positive at +0.0044 [+0.0015, +0.0073]; NorMuon versus Muon inconclusive at +0.0012 [-0.0023, +0.0047]. These are one-seed, pinned-grid results; learning rates are not independent training seeds. An inconclusive interval is not evidence of equivalence.

2 Reached weight states and retrieval prediction

At the final stage, all-rate mean cumulative hidden-weight displacement relative to weight norm is AdamW 0.004716, Muon 0.063037, NorMuon 0.063103; the corresponding last-segment ratios are AdamW 0.000355, Muon 0.006221, NorMuon 0.006190.

The declared out-of-dose predictive criterion supports: log segment/weight norm, segment effective-rank fraction, segment row-norm CV, segment top-1% row energy, cumulative/weight norm. All nine features and all four folds are retained. Undefined folds are not pooled as available cases. This is prediction outside a learning-rate dose on the same task suite, not causal mediation, statistical significance or evidence of equivalence. The exact-measurement sensitivity meets the declared predictive criterion for: Segment entropy rank. These checks do not replace the original nine-feature analysis; all five paired comparisons are reported in the appendix.

Comparator sensitivity. In post-result exploratory controls, 0 of 14 weight measurements and 0 of four functional measurements meet the predictive criterion under all four recipe-only comparators. Predictive findings depend on the comparator; we retain every baseline rather than select the one with lowest observed error. These controls neither identify a mediator nor establish equivalence.

3 Functional representation utility

Muon does not meet the joint native-coordinate criterion relative to AdamW; this does not establish equivalence. The required mean directions do not hold under every tested shared rotation. NorMuon does not meet the joint native-coordinate criterion relative to AdamW; this does not establish equivalence. The required mean directions do not hold under every tested shared rotation. With 50% of coordinates removed, relative shortlist nDCG is 0.991638/0.993778/0.992056 for AdamW/Muon/NorMuon. The figure reports all declared rates and stages.

Helpful mass share does not meet the declared held-out-dose predictive criterion. Helpful participation does not meet the declared held-out-dose predictive criterion. Degrading mass meets the declared held-out-dose predictive criterion. 50% random-removal retention does not meet the declared held-out-dose predictive criterion. This tests prediction outside a learning-rate dose on the same task suite, not causal mediation or generalization to unseen tasks.

Coordinate-level utility, its tested-basis sensitivity, and held-out-dose prediction are distinct results. Shared-rotation sign stability does not establish arbitrary-basis invariance or semantic independence of dimensions; these measurements alone do not identify a mediator of an optimizer effect.

Geometry feature	RMSE reduction	Improved	Defined	Criterion	Residual ρ
log segment/weight norm	+0.00594915	3/4	4/4	supported	-0.589
segment stable-rank fraction	-0.00112712	0/4	4/4	not supported	-0.097
segment effective-rank fraction	+0.0052098	3/4	4/4	supported	+0.608
segment row-norm CV	+0.00529116	3/4	4/4	supported	-0.597
segment top-1% row energy	+0.00488877	3/4	4/4	supported	-0.579
cumulative/weight norm	+0.0190137	4/4	4/4	supported	-0.853
cumulative stable-rank fraction	-0.00351357	0/4	4/4	not supported	-0.274
segment subspace overlap to AdamW	-0.00238356	1/4	4/4	not supported	-0.063
cumulative subspace overlap to AdamW	-0.008376	0/4	4/4	not supported	+0.035

Table 2: Every original weight-space predictor under four-fold leave-dose-index-out prediction. Decisions use exact MSE comparisons, not displayed RMSE. A missing correlation is undefined, not zero; baseline equivalence is distinct from an unidentified or numerically unresolved prediction.

Measurement	Original criterion	Exact criterion	Exact RMSE gain	Defined	Error difference
Segment stable rank	not supported	not supported	-0.00105121	4/4	+7.59091e-05
Segment entropy rank	supported	supported	+0.00558286	4/4	+0.000373052
Cumulative stable rank	not supported	not supported	-0.0035408	4/4	-2.72383e-05
Segment overlap to AdamW	not supported	not supported	-0.00159314	4/4	+0.000790415
Cumulative overlap to AdamW	not supported	not supported	-0.00406072	4/4	+0.00431528

Table 3: Every original/exact counterpart under the same four held-out learning-rate doses. RMSE gain compares the exact predictor with the nuisance-only baseline; error difference is original-predictor RMSE minus exact-predictor RMSE. Decisions use exact MSE, not rounded display values. Dashes denote undefined comparisons. These are measurement-sensitivity comparisons, not uniformly pure approximation-error tests: entropy uses the full singular spectrum, and rank/entropy averages expose their nonzero-matrix denominators. Every declared correspondence and all four folds are retained. Undefined comparisons are not available-case averages. Predictive support is not causal mediation, statistical significance, equivalence, or evidence of useful embedding dimensions.

Feature	Contrast	Effect	Simultaneous 95% CI	Decision
helpful mass share	Muon–AdamW	-0.0018016	[-0.0095111, +0.00590791]	inconclusive
helpful mass share	NorMuon–AdamW	+0.000437186	[-0.00522206, +0.00609643]	inconclusive
helpful mass share	NorMuon–Muon	+0.00223878	[-0.00674053, +0.0112181]	inconclusive
helpful participation	Muon–AdamW	+0.00358455	[+0.000143587, +0.00702552]	positive
helpful participation	NorMuon–AdamW	+0.00235085	[-0.00269661, +0.0073983]	inconclusive
helpful participation	NorMuon–Muon	-0.00123371	[-0.00668711, +0.0042197]	inconclusive
degrading mass	Muon–AdamW	-0.00130579	[-0.00396615, +0.00135456]	inconclusive
degrading mass	NorMuon–AdamW	-0.00183596	[-0.00400392, +0.00033199]	inconclusive
degrading mass	NorMuon–Muon	-0.000530169	[-0.00286867, +0.00180833]	inconclusive

Table 4: All nine final-stage co-primary contrasts, averaging every declared rate. One common-resample max- T family covers all entries; task resampling does not estimate training-seed variation.

Feature	RMSE reduction	Improved	Defined	Criterion
helpful mass share	-0.000389109	2/4	4/4	not supported
helpful participation	-0.00434356	1/4	4/4	not supported
degrading mass	+0.0100691	4/4	4/4	supported
50% random-removal retention	+0.000341663	2/4	4/4	not supported

Table 5: All four functional predictors beyond optimizer, stage and within-optimizer dose. Decisions use exact MSE comparisons; displayed RMSE is not a significance test. Undefined folds are retained, never pooled as available cases.

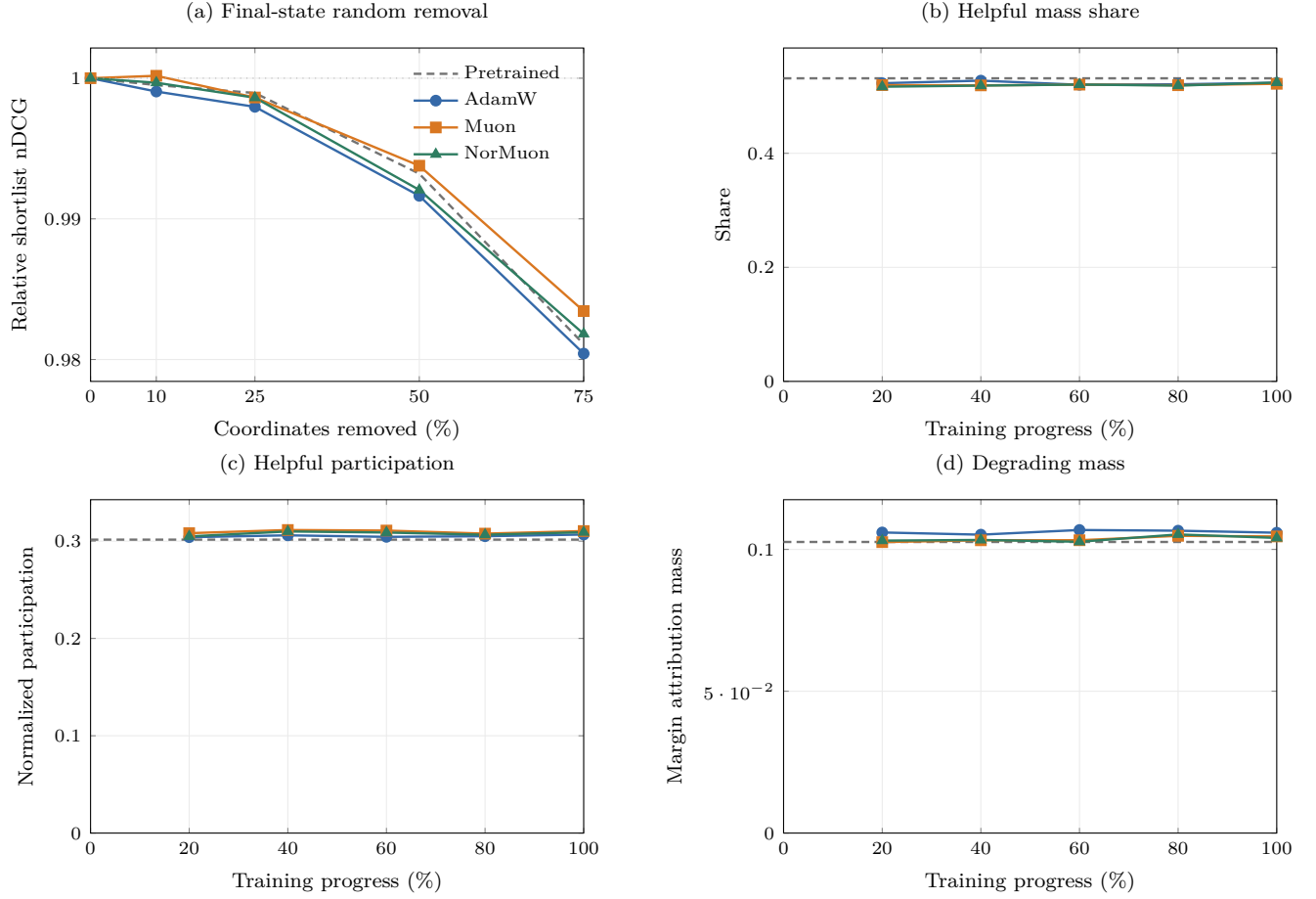


Figure 1: Dimension utility over the complete rate grid. (a) Final-state retention after shared random coordinate removal; the zero-removal value is one by definition. (b–d) Native-coordinate margin attribution at all five stages. Each optimizer curve averages all four rates and 14 tasks; (a) additionally averages 20 shared masks per nonzero fraction. The gray curve or horizontal line is the pretrained reference. Lines join measured points, not fitted trajectories. These are descriptive means, not confidence intervals or full-corpus retrieval scores; formal contrasts and rotation controls are reported separately.

4 All post-result recipe comparisons

The four recipe-only comparators retain the original outcome and held-dose folds. B0 contains optimizer indicators, stage indicators and within-optimizer centered log learning rate. B1 adds optimizer-specific raw learning rate. B2 uses optimizer-specific stage indicators and log/raw rate slopes. B3 adds the nominal accumulated learning-rate schedule. The latter is not a measured weight-update norm. B0+D and B3+D additionally condition on cumulative weight displacement. Each cell adds just one measurement, and all four folds remain.

Measurement	B0	B1	B2	B3	B0+D	B3+D
Comparator alone	2.905	0.652	1.391	1.439	1.004	1.420
log segment/weight norm	2.310* (3/4)	0.687 (2/4)	1.357* (4/4)	1.393* (4/4)	1.247 (1/4)	1.376* (4/4)
segment stable-rank fraction	3.018 (0/4)	0.524* (3/4)	1.435 (3/4)	1.597 (0/4)	1.050 (2/4)	1.557 (0/4)
segment effective-rank fraction	2.384* (3/4)	0.676 (2/4)	1.378* (3/4)	1.435* (3/4)	1.221 (1/4)	1.418* (3/4)
segment row-norm CV	2.376* (3/4)	0.630* (3/4)	1.401 (1/4)	1.448 (0/4)	1.142 (1/4)	1.428 (1/4)
segment top-1% row energy	2.416* (3/4)	0.559* (3/4)	1.396 (1/4)	1.449 (1/4)	1.032 (2/4)	1.429 (1/4)
cumulative/weight norm	1.004* (4/4)	0.795 (1/4)	1.407 (2/4)	1.420* (4/4)	1.004 (0/4)	1.420 (0/4)
cumulative stable-rank fraction	3.257 (0/4)	1.070 (2/4)	1.442 (1/4)	1.561 (1/4)	1.244 (1/4)	1.562 (1/4)
segment subspace overlap to AdamW	3.144 (1/4)	0.651 (2/4)	1.539 (0/4)	1.676 (0/4)	0.816* (3/4)	1.645 (0/4)
cumulative subspace overlap to AdamW	3.743 (0/4)	1.020 (1/4)	1.623 (2/4)	1.924 (1/4)	1.269 (1/4)	1.335 (2/4)
Exact: Segment stable rank	3.010 (0/4)	0.528* (3/4)	1.438 (3/4)	1.625 (0/4)	1.057 (2/4)	1.582 (0/4)
Full-spectrum segment entropy	2.347* (3/4)	0.689 (2/4)	1.294* (4/4)	1.356* (4/4)	1.260 (2/4)	1.348* (4/4)
Exact: Cumulative stable rank	3.259 (0/4)	1.001 (2/4)	1.313* (4/4)	1.373* (3/4)	1.213 (0/4)	1.380 (2/4)
Exact: Segment overlap to AdamW	3.065 (1/4)	0.601 (2/4)	1.507 (1/4)	1.645 (0/4)	0.840* (3/4)	1.596 (0/4)
Exact: Cumulative overlap to AdamW	3.311 (2/4)	1.398 (1/4)	1.457 (0/4)	2.140 (0/4)	1.452 (1/4)	2.017 (0/4)

Table 6: Complete post-result exploratory recipe-control sensitivity. Cells show pooled held-dose RMSE in nDCG@10 points (lower is better), and the number of improved folds. An asterisk marks strictly lower exact pooled MSE and improvement in at least three of four folds; it is not a significance test. D is cumulative weight displacement. All six comparator variants are retained.

Measurement	B0	B1	B2	B3	B0+D	B3+D
Comparator alone	2.905	0.652	1.391	1.439	1.004	1.420
Helpful mass share	2.944 (2/4)	0.634 (2/4)	1.398 (1/4)	1.481 (1/4)	1.033 (0/4)	1.466 (1/4)
Helpful participation	3.340 (1/4)	0.891 (0/4)	1.552 (3/4)	1.422* (4/4)	1.062 (2/4)	1.428 (1/4)
Degrading mass	1.898* (4/4)	0.623 (2/4)	1.424 (1/4)	1.508 (0/4)	0.907* (3/4)	1.482 (1/4)
50% removal retention	2.871 (2/4)	0.600* (3/4)	1.451 (2/4)	1.476 (1/4)	1.002 (2/4)	1.452 (0/4)

Table 7: Complete post-result exploratory recipe-control sensitivity. Cells show pooled held-dose RMSE in nDCG@10 points (lower is better), and the number of improved folds. An asterisk marks strictly lower exact pooled MSE and improvement in at least three of four folds; it is not a significance test. D is cumulative weight displacement. All six comparator variants are retained.